LEFT DUALS IN THE MONOIDAL CATEGORY OF COMPLEXES ARE THE TERMWISE DUALS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Uniqueness of left duals. Categories, Remark categories-remark-left-dual-adjoint. Left duals are unique up to unique isomorphism.
- Module-level left duals. Modules, Lemma modules-lemma-left-dual-module. F^n is locally a direct summand of a finite free module.

Source reference. Original reference 0FFP: sources/stacks-chow/tags.txt, source lines 19294–19294. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0FP9: sources/stacks-chow/tags.txt, source lines 19594–19594. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0FFR: sources/stacks-chow/tags.txt, source lines 19296–19296. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0FNW: sources/stacks-chow/tags.txt, source lines 19581–19581. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}^\bullet$ be a complex of $\mathcal{O}_X$-modules. If $\mathcal{F}^\bullet$ has a left dual in the monoidal category of complexes of $\mathcal{O}_X$-modules (Categories, Definition 0FFP) then $\mathcal{F}^\bullet$ is a locally bounded complex whose terms are locally direct summands of finite free $\mathcal{O}_X$-modules and the left dual is as constructed in Example 0FP9.*

Proof. By uniqueness of left duals (Categories, Remark 0FFR) we get the final statement provided we show that $\mathcal{F}^\bullet$ is as stated. Let $\mathcal{G}^\bullet, \eta, \epsilon$ be a left dual. Write $\eta = \sum \eta_n$ and $\epsilon = \sum \epsilon_n$ where $\eta_n : \mathcal{O}_X \to \mathcal{F}^n \otimes_{\mathcal{O}_X} \mathcal{G}^{-n}$ and $\epsilon_n : \mathcal{G}^{-n} \otimes_{\mathcal{O}_X} \mathcal{F}^n \to \mathcal{O}_X$. Since $(1 \otimes \epsilon) \circ (\eta \otimes 1) = \mathrm{id}_{\mathcal{F}^\bullet}$ and $(\epsilon \otimes 1) \circ (1 \otimes \eta) = \mathrm{id}_{\mathcal{G}^\bullet}$ by Categories, Definition 0FFP we see immediately that we have $(1 \otimes \epsilon_n) \circ (\eta_n \otimes 1) = \mathrm{id}_{\mathcal{F}^n}$ and $(\epsilon_n \otimes 1) \circ (1 \otimes \eta_n) = \mathrm{id}_{\mathcal{G}^{-n}}$. Hence we see that $\mathcal{F}^n$ is locally a direct summand of a finite free $\mathcal{O}_X$-module by Modules, Lemma 0FNW. Since the sum $\eta = \sum \eta_n$ is locally finite, we conclude that $\mathcal{F}^\bullet$ is locally bounded. □

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